

None

pandas.wide_to_long

We use `pd.wide_to_long` to reshape the DataFrame from wide format to long format.

Sales DataFrame:

	Store	sales_2020	sales_2021
0	A	100	150
1	B	90	120



reshaped_sales_data:

sales

Store	Year	sales
A	2020	100
B	2020	90
A	2021	150
B	2021	120

```
#Import Libraries
import pandas as pd, numpy as np
```

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Thank You!

1st Edition

Why This E-book?

"The aim of this ebook is to give you the 'aha' moment right away at the start of learning a new concept."

- Practical step By Step Guide With Simple Examples
- Visual Illustrations and Interactive
- Simple Datasets
- Comprehensive Coverage (pandas Documentation used as reference)

Tips for Effective Use of This Ebook

Use Examples First: If you don't understand the text, go straight to the examples—they're self-explanatory. After reviewing them, return to the text to grasp the practical concepts.

Practice with Datasets: Copy the datasets provided and practice using them to reinforce your understanding.

Click on the blue links to go directly to the section you want to learn about.

Introduction

We use `pd.wide_to_long` to reshape the DataFrame from wide format to long format.

Sales DataFrame:

	Store	sales_2020	sales_2021
0	A	100	150
1	B	90	120



reshaped_sales_data:

		sales
Store	Year	
A	2020	100
B	2020	90
A	2021	150
B	2021	120

The `df` parameter

The "df" parameter represents the DataFrame that is in wide format.

The `stubnames` parameter

The "stubnames" parameter, which can be a string or a list of strings, refers to the base name(s) of the columns. It is assumed that the wide-format variables start with these base names.

Output:

Code Snippet:

Single Stub Name: sales_df

Here, "sales" is the stub name,

```
pd.wide_to_long(
    df = sales_df,
```

and it tells pandas that columns starting with sales_ should be unpivoted.

pandas.wide_to_long

	Store	sales_2020	sales_2021
0	A	100	150
1	B	90	120

→

	Store	Year	sales
	A	2020	100
	B	2020	90
	A	2021	150
	B	2021	120

```
stubnames = 'sales',
i = 'Store',
j = 'Year',
sep = '_'
)
```

Code Snippet:

```
data = {
    'Store': ['A', 'B'],
    'sales_2020': [100, 90],
    'sales_2021': [150, 120]
}

sales_df = pd.DataFrame(data)
sales_df
```

sales_df:

	Store	sales_2020	sales_2021
0	A	100	150
1	B	90	120

Reshaping with Multiple Stub Names

Code Snippet:

df

	Store	sales_2020	sales_2021	expenses_2020	expenses_2021
0	A	100	150	80	120
1	B	90	120	60	90

Output:

	Store	Year	sales	expenses
	A	2020	100	80
	B	2020	90	60
	A	2021	150	120
	B	2021	120	90

```
pd.wide_to_long(
    df = df,
    stubnames=['sales',
               'expenses'],
    i = 'Store',
    j = 'Year',
    sep = '_'
)
```

Code Snippet:

```
df = pd.DataFrame({
    'Store': ['A', 'B'],
    'sales_2020': [100, 90],
    'sales_2021': [150, 120],
    'expenses_2020': [80, 60],
    'expenses_2021': [120, 90]
})
```

df

df:

	Store	sales_2020	sales_2021	expenses_2020	expenses_2021
0	A	100	150	80	120
1	B	90	120	60	90

The **i** parameter

The parameter "i" specifies the column(s) that should be used as identifier variable(s). This means that the values in these columns will be retained as they are, while the data in the wide-format columns is transformed into a long format.

It can be either a single string or a list-like structure.

When "i" is a string,
it indicates a single column to be
used as an identifier

sales_df

	Store	sales_2020	sales_2021
0	A	100	150
1	B	90	120



Output:

		sales
Store	Year	
A	2020	100
B	2020	90
A	2021	150
B	2021	120

Code Snippet:

```
pd.wide_to_long(  
    df = sales_df,  
    stubnames = 'sales',  
    i = 'Store',  
    j = 'Year',  
    sep = '_'  
)
```

Code Snippet:

```
data = {
    'Store': ['A', 'B'],
    'sales_2020': [100, 90],
    'sales_2021': [150, 120]
}

sales_df = pd.DataFrame(data)
sales_df
```



df:

	Store	sales_2020	sales_2021	expenses_2020	expenses_2021
0	A	100	150	80	120
1	B	90	120	60	90

When "i" is list-like,

it indicates multiple columns that collectively serve as identifier variables.

df

	Store	Region	sales_2020	sales_2021
0	A	North	100	150
1	B	South	90	120



Output:

	Store	Region	year	sales
A	North		2020	100
			2021	150
B	South		2020	90
			2021	120

Code Snippet:

```
pd.wide_to_long(
    df = df,
    stubnames = 'sales',
    i = ['Store', 'Region'],
    j = 'year',
    sep = '_'
)
```

Code Snippet:


```
df = pd.DataFrame({
    'Store': ['A', 'B'],
    'Region': ['North', 'South'],
    'sales_2020': [100, 90],
    'sales_2021': [150, 120]
})
df
```

→

df:

	Store	Region	sales_2020	sales_2021
0	A	North	100	150
1	B	South	90	120

The `j` parameter

The "j" parameter is used to specify the name for the sub-observation variable in the long format. It allows you to define the suffix you want to use for the variable names in the transformed data.

The dtype of "j" is str (string).

The `j` parameter specifies the name for the sub-observation variable.

In the example, `j="Year"` is used to

Sales DataFrame:

Output:

	Store	Year	sales
	A	2020	100

Code Snippet:

```
pd.wide_to_long(
    df = df,
    stubnames = 'sales',
```

name the suffixes
(e.g., 2020, 2021)
when
transforming
data from wide
format to long
format.

	Store	sales_2020	sales_2021
0	A	100	150
1	B	90	120



	Store	Year	sales
	B	2020	90
	A	2021	150
	B	2021	120

```
i = 'Store',
j = 'Year',
sep = '_'
)
```

Code Snippet:

```
data = {
    'Store': ['A', 'B'],
    'sales_2020': [100, 90],
    'sales_2021': [150, 120]
}
```

```
df = pd.DataFrame(data)
```

```
df
```

df:

	Store	sales_2020	sales_2021
0	A	100	150
1	B	90	120

The **sep** parameter

The "sep" parameter is a string, and its default value is an empty string ("").

A character indicating the separation of the variable names in the wide format, to be stripped from the names in the long format. For example, if your column names are A-suffix1, and A-suffix2, you can strip the hyphen by specifying sep="-".

Sales DataFrame:

	Store	sales_2020	sales_2021
0	A	100	150
1	B	90	120



Output:

	Store	Year	sales
	A	2020	100
	B	2020	90
	A	2021	150
	B	2021	120

Code Snippet:

```
pd.wide_to_long(
    df = df,
    stubnames = 'sales',
    i = 'Store',
    j = 'Year',
    sep = '_'
)
```

The `suffix` parameter

The `suffix` parameter is used to specify which part of the column names should be treated as suffixes when converting data from wide to long format.

Default Value "\d+":
By default, suffix="\d+" captures numeric suffixes to be stripped from the names in the long format.

Sales DataFrame:

	Store	sales_2020	sales_2021
0	A	100	150
1	B	90	120



Output:

	Store	Year	sales
	A	2020	100
	B	2020	90
	A	2021	150
	B	2021	120

Code Snippet:

```
pd.wide_to_long(
    df,
    stubnames='sales',
    i = 'Store',
    j = 'Year',
    sep = '_',
    suffix='\\d+'
)
```

Use "\\D+" to capture suffixes with letters or other non-numeric characters.

df:

	Store	sales_YearOne	sales_YearTwo
0	A	100	150
1	B	90	120



Output:

	Store	Year	sales
	A	YearOne	100
	B	YearOne	90
	A	YearTwo	150
	B	YearTwo	120

Code Snippet:

```
pd.wide_to_long(
    df,
    stubnames='sales',
    i = 'Store',
    j = 'Year',
    sep = '_',
    suffix='\\D+'
)
```

Use "\D+" to

capture
suffixes

with letters
or other
non-
numeric
characters.

df:

	Store	sales_y1	sales_y2	sales_y3
0	A	100	150	130
1	B	90	120	80

→

Output:

	Store	Year	sales_y3	sales
	A	y1	130	100
	B	y1	80	90
	A	y2	130	150
	B	y2	80	120

Code Snippet:

```
pd.wide_to_long(
    df,
    stubnames='sales',
    i='Store',
    j='Year',
    sep='_',
    suffix='(!?y1|y2)'
)
```

Sources & References

[pandas.wide_to_long Documentation](https://pandas.pydata.org/docs/reference/api/pandas.wide_to_long.html)

(https://pandas.pydata.org/docs/reference/api/pandas.wide_to_long.html)

Author Biography

Kichere Magubu is a data enthusiast and content creator.

Other E-books

[Download](https://github.com/kicherethedatascientist/My_e-Books) (https://github.com/kicherethedatascientist/My_e-Books)

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Thank You!

```
In [30... #print("The cell to convert jupyter notebook to html")
!jupyter nbconvert --to hide_code_html "pandas.wide_to_long.ipynb"
```

```
[NbConvertApp] Converting notebook pandas.wide_to_long.ipynb to hide_code_html
[NbConvertApp] Writing 641732 bytes to pandas.wide_to_long.html
```

```
In [ ]:
```